Udit Agarwal

Applied AI Engineer | Forward Deployed Engineer | Full-Stack and Backend Software Engineer

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SUMMARY

Software engineer with 3+ years of combined full-time and internship experience building backend systems, full-stack applications, and LLM-powered features for client and product teams. Ships RAG and structured-output features with evaluation and guardrails, multi-tenant APIs with role-based access control, and data integrations in Python and TypeScript, from client discovery through tested, CI-gated cloud deployment.

TECHNICAL SKILLS

Languages and Frontend: Python, TypeScript, JavaScript, SQL, React, Next.js, Tailwind CSS

Applied AI and LLM Engineering: RAG, Vector Search (pgvector, FAISS), Structured Outputs, LLM Evaluation, Guardrails, Gemini API, OpenAI API, Vercel AI SDK

Backend, APIs and Architecture: FastAPI, Django REST Framework, Node.js, Fastify, REST APIs, Webhooks, Multi-Tenant Architecture, JWT, Role-Based Access Control (RBAC)

Databases and ML: PostgreSQL, Prisma, MongoDB, Redis, TensorFlow/Keras, ONNX Runtime, Model Evaluation, Data Leakage Auditing

Cloud, Testing and Delivery: AWS (EC2, ECS, S3, RDS, Lambda), Docker, GitHub Actions (CI/CD), Vercel, Render, Firebase, Hugging Face, pytest, Vitest

Certifications: Oracle Certified Associate – Agentic AI (2026), DeepLearning.AI Deep Learning (Neural Networks, Optimization), AWS Cloud Foundations

EXPERIENCE

Merkle | ***Software Engineer, Applied AI (Client Solutions)*** Jun 2026 – Present

- Built a cited, access-controlled RAG assistant over CRM notes, campaign briefs, and analytics reports using PostgreSQL and pgvector (HNSW), tracking retrieval quality against labeled questions
- Implemented production-safe LLM inquiry routing in FastAPI using Pydantic-validated structured outputs and a rules-based fallback for low-confidence responses
- Integrated client CRM and marketing platforms into a shared data layer without duplicate records via REST APIs and webhooks with per-client field mappings, idempotency keys, and retry with backoff
- Blocked prompt and model regressions from reaching production by enforcing CI evaluation gates across accuracy, latency, and cost
- Led technical discovery with client stakeholders, turned their requirements into API contracts, and delivered a React and TypeScript review console on AWS ECS with PII redaction and per-client spend caps

Mindfire Solutions | ***Software Engineer*** Jan 2023 – Jul 2024

- Built the backend for a healthcare client workflow platform in Django REST Framework and PostgreSQL, with role-based permissions, audit logging, and versioned REST APIs for web and mobile clients
- Accelerated slow reports using EXPLAIN ANALYZE profiling, composite indexes, and nightly materialized views for heavy aggregations
- Eliminated duplicate charges from replayed payment events by integrating payment and SMS gateways through signed webhooks, idempotent handlers, and Celery and Redis retry queues
- Prototyped question-answering over client PDFs with OpenAI embeddings, FAISS, and FastAPI, and presented accuracy and cost trade-offs

Mindfire Solutions | ***Software Engineering Intern*** Jan 2022 – Dec 2022

- Shipped Django REST Framework endpoints and serializers into existing client applications through senior engineer code review
- Fixed QA-reported defects reproduced from application logs, covering each fix with pytest tests, and wrote PostgreSQL migrations

PROJECTS

RxFlow - Multi-Tenant Pharmacy ERP | *Next.js, TypeScript, Fastify, Prisma, PostgreSQL, Turborepo, GitHub Actions*

- Architected a multi-tenant pharmacy ERP spanning 133 API routes across 22 modules as a pnpm/Turborepo monorepo (Fastify, Next.js, shared Prisma), isolating every tenant by scoping each query to the JWT and enforcing one 8-role access matrix across 124 authenticated routes
- Made the ERP deployable in any country with zero code changes, supporting 5 regional profiles and 4 pluggable tax engines (India GST, VAT, US sales tax, none), by moving currency, timezone, fiscal year, tax-ID validation and drug-regulation rules into a shared configuration package
- Closed a critical cross-tenant data-exposure path, where replayed refresh tokens silently dropped Prisma's tenant filter, and eliminated overselling under concurrent billing by enforcing stock checks inside the update itself, verified by 82 Vitest tests and scripted end-to-end runs

PrepWise - AI Voice Interview Platform | *Next.js, TypeScript, Gemini, Vercel AI SDK, Zod, Web Speech API, Firebase*

- Built an adaptive interviewer that dynamically adjusts to candidate responses (dropping confidence scores from 50 to 20 on hesitant answers to trigger fundamentals probes) using 1 schema-validated LLM call per turn for grading, state updates, and question selection
- Raised free AI capacity from 20 to 1,000 requests/day and cut question generation from ~8-10s to ~2.5s with a Gemini Flash-Lite model chain that fails over on quota exhaustion; cut speech cost to $0 with browser-native recognition
- Bounded the agent loop with code-enforced guards the model can't override (12-turn hard cap, 3-turn cap per topic) and shipped in-browser JavaScript and Python coding rounds, covered by 172 Vitest tests in CI

NeuroClass - Brain Tumor MRI Classifier (IEEE 2024) | *Python, TensorFlow/Keras, ONNX Runtime, FastAPI, Docker, Hugging Face*

- Co-authored an IEEE ICC-ROBINS 2024 paper on brain tumor MRI classification, then audited its benchmark and exposed train/test leakage by comparing 8.96M image pairs with perceptual hashing (97.2% vs 86.5% accuracy on near-duplicate vs novel scans), and traced a hidden 2.7-point accuracy gap to GPU numerics by pinning evaluation to the deployed CPU path
- Deployed the classifier on Hugging Face Spaces with ONNX Runtime, FastAPI and Docker at 91.25% accuracy and 0.980 macro ROC-AUC, cutting API latency 5.7x (420 to 74 ms) by compiling Grad-CAM and moving inference off the event loop, and calibration error 63% with temperature scaling, gated by 35 pytest tests in CI

EDUCATION

Indiana University Bloomington Bloomington, Indiana, US

Master of Science in Computer Science, GPA: 3.84/4.0 Aug 2024 – May 2026

Kalinga Institute of Industrial Technology (KIIT University) Bhubaneswar, Odisha, India

Bachelor of Technology in Computer Science and Engineering, GPA: 8.85/10.0 Aug 2020 – May 2024